U1-function/mapping

Input → output is called mapping{

Algebraically(linear): $x \mapsto x + 1$

Read as: x is mapped to x+1
}{one-one (one input one output) **injective** /e.g. f(x)=2x+1
many-one (more input one output) /e.g. f(x)=x^2
one-many (one input more output) /e.g. $f(x) = \pm\sqrt{x}$

U1-function/definition of function

-{maps each x value to just one y value for a defined set of input value.

one-one, more-one is called function}

{it can be written as:

$$f: x \mapsto x + 1 \quad x \in R$$
$$f(x) = x + 1 \quad x \in R$$

Read as: the function f, such that x is mapped to x+1

/f(x) represents the output values for the function f.

}

U1-function/composite functions

A composite function is a function that is formed by applying one function to the result of another function.
If you have two functions, f and g, the composite function $f \circ g$
Read as "f composed with g" is defined as

$$(\mathrm{f} \circ \mathrm{g})(\mathrm{x}) = \mathrm{f}(\mathrm{g}(\mathrm{x}))$$

This means you first apply function g to function x, then apply f to the result.

/function f is outer function; g is inner function;

(/ in IG 0606 ADD, we don't use $f \circ g$, we use fg(x);
/e.g.: domain(gf) = {x ∈ domain f | f(x) ∈ domain g}

}

U1-function/modulus functions

-*:'modulus' means 'absolute';
-{f(x)=$|x|$ is modulus functions}

U1-function/graphs of y=|f(x)| where f(x) in linear

{step:
1. Draw the graph of linear function

2.then reflect in the x-axis the part of the line that below the x-axis

}

U1-function/inverse function

-{In f^{-1}(x)} Swap the order of domain and range.}

/EXAM:A symmetrical even function must not have an inverse function. (Unless the domain is restricted, only one-one functions have inverse function)

U2-simultaneous equations/one linear and one non-linear

-U2 {key words:
parabola: type of curve such as the path of something that is thrown up then comes down.

turning point: a point at which significant change occurs.

stationary point: point is on the function graph where the derivative is zero (rate of change=0);$f'(x) = 0$
discriminant/ cn: 判別式
| stationary point = turning point in function in graph.

}{Solve- simultaneous equation: 1 linear and 1 non-linear{

$y=x^2-5$ & $y=x+1$

$x+1 = x^2-5$

$(x+2)(x-3)=0$

$x=-2$ or 3

}

U2-simultaneous equations/maximum & minimum values of a quadratic function

Quadratic function:$y = ax^2 + bx + c$ (a ≠ 0)

The graph of quadratic function is parabola （cubic function is not parabola）

{ If concave down ($a < 0$); curve has maximum point

If concave up ($a > 0$); curve has minimum point

the maximum or minimum point is called : turning point ('center' of concave) OR stationary point (rate of change = 0)

}Completing the square of quadratic function{

name	formula	information	Use-time
Simple form	$y = ax^2 + bx + c$	**y-intercept** (0, c) concave rise/down （a）	3 points
vertex form	$y = a(x - h)^2 + k$ -b/2a : vertex x $vertex: y = \frac{4ac-b^2}{4a}$ $y = f(\frac{-b}{2a})$=a$(\frac{-b}{2a})$+b$\left(\frac{-b}{2a}\right)$+c	vertex (h, k) symmetry x = h **extreme value** k	Given the vertex and another point, find the max/min value
Factorized form	$y = a(x - p)(x - q)$	x-axis (p, 0)and(q, 0)	x-intercept

1. extract leading cofficient：$y = 2(x^2 - 4x) + 5$。 $//x^2 - 4x$
2. 配方：目标是让括号里的 $x^2 - 4x$ 变成 $(x - m)^2$ 的形式。
   - 我们知道 $(x - m)^2 = x^2 - 2mx + m^2$。
   - 对比 $x^2 - 4x$, 可知 $-2m = -4$, 所以 $m = 2$。
   - 要得到 $(x - 2)^2$, 我们需要 $x^2 - 4x + 4$。但原式是 $x^2 - 4x$,
   - $x^2 - 4x = (x^2 - 4x + 4) - 4 = (x - 2)^2 - 4$
3. **Substitute** ：$y = 2[(x - 2)^2 - 4] + 5$
4. **simplify**：$y = 2(x - 2)^2 - 8 + 5 = 2(x - 2)^2 - 3$
5. **Obtain the vertex form**：$y = 2(x - 2)^2 + (-3) \rightarrow$ max/min point $(2, -3)$, symmetry $x = 2$, min value -3。

}**Example 1: y=2x2+8x+5{**

1. **Factor out the leading coefficient from the first two terms**

   $y=2(x^2+4x)+5$

2. **Complete the square inside the parentheses**
   For x^2+4x
   Half of the coefficient of xx is 4÷2=2. Square it: $2^2=4$
   Add and subtract 4:

$x^2+4x=(x^2+4x+4)-4=(x+2)^2-4$

3. **Substitute back**

   $y=2[(x+2)^2-4]+5$

4. **Simplify**

   $y=2(x+2)^2-8+5$
   $y=2(x+2)^2-3$

Vertex form: $y=2(x+2)^2-3$
Vertex: $(-2,-3)$

}

Simple form to quadratic radical square root expression{
1. write down the function
2. use discriminant to calculate and determine the root.{判断是否可行}
3. use $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$ to get the root and simplify. {like: $x=\frac{8\pm4}{4}$ get x=1 & 3}
4. write down the factorized form(or intercept form) (different or same, if the function isn't throuth or reflect in the zero point, we can't use the way to calculate)
}vertex form to quadratic radical square root expression(solve the root){
1. Let y=0, solve the function
2. extraction of square root
3. write down the factorized form (intercept form)
{e,g.

$$2(x-2)^2-2=0$$
$$(\mathrm{x}-2)^2=1$$
$$x-2=\pm1$$
$$y=2(x-3)(x-1)$$

}

U2-simultaneous equations/ graphs of y=|f(x)| where f(x) in quadratic

-{ form: $y = |ax^2 + bx + c|$}

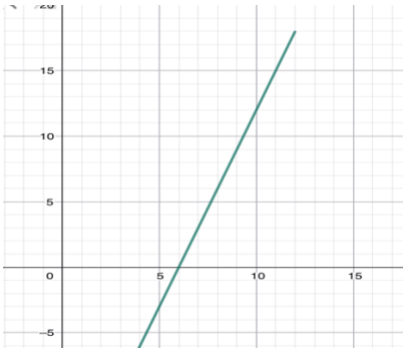
-{step: make the region which under the x-axis to the region which above the x-axis, from through the zero point(s) to reflect in the zero point(s).}

U2-simultaneous equations/quadratic equality

{solve as equality:
e.g.: $2x-5<9$ (course book is wrong: P40)

$$2x < 14$$

$$x < 7$$



The region of x< 12 is 'not shown'
}

U2-simultaneous equations/roots of quadratic function & Intersection of a line and a curve

{solve the x in the quadratic function

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ = $x = \frac{-b \pm \sqrt{\Delta}}{2a}$;

$\Delta = b^2 - 4ac$ is discriminant}

b^2-4ac	roots	Line and curve with expression
>0	2 real distinct roots	2 distinct points of intersection; the function through the zero points. /expression: the line cuts the curve at 2 distinct point
=0	2 equal real roots	1 points of intersection, the function reflect in the zero point /expression: the line touches the curve at one point, this means that the line is a tangent to the curve.
<0	0 real roots	No point intersection; no zero point. /expression: the line does not intersect the curve

/the condition for a quadratic equation to have (a) real root(s) is $b^2 - 4ac \geq 0$

{factories with delta (discriminant): $5(x^2 - 14x - 3)$

$$\Delta = b^2 \pm 4ac = (-14)2 - 4 \cdot 1 \cdot (-3) = 196 + 12 = 208$$

$$\sqrt{208} = \sqrt{(16 \times 13)} = 4\sqrt{13}$$

$$x2 - 14x - 3 = \frac{\left(x - 14 + 4\sqrt{13}\right)}{2} \times \frac{(x - 14 - 4\sqrt{13})}{2}$$

$$= (\mathrm{x} - (7 + 2\sqrt{13}))(\mathrm{x} - (7 - 2\sqrt{13}))$$

$$5(x2 - 14x - 3) = 5(x - 7 - 2\sqrt{13})(x - 7 + 2\sqrt{13})$$

*if we can't(or hard) get the factor in the integer.

}

U3-Factors and polynomial-introduction

-{form of polynomial: $a^n x^n + a^{n-1} x^{n-1} + \cdots + a^1 x + a^0$}

U3-Factors and polynomial/division of polynomial & factor theorem & remainder theorem

When we use a expression of first degree (x-k) to divide polynomial P(x), quotient Q(x) is a polynomial of (n-1), remainder R is constant number.

$$P(x) = (x - k) \cdot Q(x) + R$$

Long Division: (example:$2x^3 - 5x^2 + 3\mathrm{x} - 4$

| $2x^2 - x + 1.$

x-2. | $2x^3 - 5x^2 + 3\mathrm{x} - 4$

| $2x^3 - 4x^2$

|. $-x^2+3x$

|. x^2+2x

| x-4

| x-2

|. 2

}RULE:
排列被除式和除式，按降幂排列，缺项补 0。

Arrange the dividend and divisor in descending powers of x, filling any missing terms with 00.

用除式的首项去除被除式的首项，得到商的第一项，写在商的位置。

Divide the first term of the dividend by the first term of the divisor to get the first term of the quotient. Write it above the division bar.

将商的该项乘以整个除式，将结果写在被除式下面，对齐同类项，然后相减。

Multiply this quotient term by the entire divisor, write the product below the dividend (aligning like terms), and subtract.

落下被除式的下一项，与余式组成新的被除式。

Bring down the next term of the original dividend to form a new polynomial.

重复，直到余式的次数低于除式的次数。

Repeat steps until the remainder has a lower degree than the divisor.

} Remainder Theorem{

When a polynomial P(x) is divided by (x−r), the remainder is R=R(r).
That is:

$$P(x) = (x - r) \cdot Q(x) + R$$

$$P(x) = R$$

Remainder Theorem Techniques / 余数定理技巧

Technique	中文	English
1	若除式为 ax+b (a≠1)，先化为 x+b/a 形式，余数为 P(−b/a)	If the divisor is ax+bax+b (a≠1), rewrite as x+b/a; the remainder is P(−b/a)
2	利用余数定理求未知系数：根据已知余数列方程求解。	Use the Remainder Theorem to find unknown coefficients: set up an equation from the given remainder.
3	余数定理与因式定理结合：若余数 P(k)=0，则 x−k 是因式。	Combine with the Factor Theorem: if the remainder P(k)=0, then x−k is a factor.
4	对于多项式除法，若只需求余数，不必做整步除法，直接代入即可。	If only the remainder is needed, avoid full division; just evaluate P(k).

}Factor Theorem:

$$If\ P(r) = 0, then\ x - r\ is\ a\ factor\ of\ P(x)$$
$$That\ is: P(x) = (x - r) \cdot Q(x), and\ the\ remainder\ is\ 0.$$

} synthetic division{

Synthetic Division

$(A\ faster\ alternative\ to\ long\ division\ when\ the\ divisor\ is\ of\ the\ form\ x-k)$

$\boldsymbol{Example}$: $Divide\ P(x) = 2x^3 - 5x^2 + 3x - 4\ by\ x - 2.$

1. $Write\ the\ coefficients\ of\ P(x)\ 2, -5, 3, -4$
2. $Write\ k = 2\ on\ the\ left$
3. $Bring\ down\ the\ first\ coefficient\ 2$
4. $Multiply\ by\ k\ (2 \times 2 = 4), add\ to\ the\ next\ coefficient:\ -5 + 4 = -1 Multiply\ by\ k\ (-1 \times 2 = -2), add\ to\ the\ next\ coefficient: 3 + (-2) = 1 Multiply\ by\ k\ (1 \times 2 = 2), add\ to\ the\ last\ coefficient:\ -4 + 2 = -2\ (remainder)$

$\boldsymbol{Result}$:

$Quotient\ Q(x) = 2x^2 - x + 1$

$Remainder\ R = -2$

*remainder * the divisor = subtrahend of next step
*last number is “finial remainder”

}advanced for testing root{

$Advanced\ Techniques\ for\ Factoring\ Polynomials$

$Using\ the\ Factor\ Theorem\ to\ Find\ Roots, Then\ Reduce\ the\ Degree$

$For\ a\ cubic\ polynomial\ \left(P(x)\right)$:

Try possible rational roots:

$$\pm \frac{factor\ of\ constant}{factor\ of\ leading\ coefficient factor}$$

(e.g., x=±1,±2,±12,...)

Find r such that P(r)= 0: then x−r is a factor.

Divide P(x) by x−r using long division or synthetic division to obtain a quadratic quotient.

Factor the quadratic (by inspection, cross-multiplication, or the quadratic formula).

\e.g.:
Example: $P(x)=x^3-6x^2+11x-6$

Try x=1: P(1)=1−6+11−6=0⇒ x−1 is a factor.

Synthetic division yields quotient x^2-5x+6

Factor the quadratic: x2−5x+6=(x−2)(x−3)

Final factorization: P(x)=(x−1)(x−2)(x−3)

Step	Action
1. **Rational Root Test**	List possible rational roots: $\pm\frac{factor\ of\ constant}{factor\ of\ leading\ coefficient factor}$ Try them in P(x) until P(r)=0
2. **Factor Theorem**	If P(r)=0, then x−r is a factor.
3. **Polynomial Division**	Divide P(x)Pby x−r using long division or synthetic division to get the quotient Q(x) (degree one less).
4. **Continue Factoring**	If Q(x) is quadratic, factor it (cross-multiplication or quadratic formula). If Q(x) has degree ≥ 3, repeat steps 1–3.
5. **Final Form**	Write P(x)=(x−r1)(x−r2)⋯ (include the leading coefficient if necessary).

}

U3-Factors and polynomial/Cubic expression and equation

-{form: f(x)= $f(x) = ax^3 + bx^2 + cx + d$}

/also can use division of polynomial & factor theorem & remainder theorem

{Remainder Theorem – Example

Find the remainder when $\mathrm{P(x)} = 2x^2 - 5x^2 + 3\mathrm{x} - 4$ is divided by x−2.

$$P(2) = 2(8) - 5(4) + 3(2) - 4 = 16 - 20 + 6 - 4 = -2$$

The remainder is −2.

}

U4-Equations, inequalities and graphs/ solving equations of the type | ax+b | = | cx+d |

{ |p| = |q| ⇔ $p^2 = q^2$

$ax + b = cx + d.\mathbf{OR}\ ax + b = -(cx + d)$; It indicates that the absolute values of two

expressions are equal. This is equivalent to two expressions being equal or opposite to each other.
only solve 2 linear equations, then merge all the solutions

/ The graph of each absolute value function takes the shape of a "V".
/ The number of intersection points of two "V" shapes may be 0, 1, 2 or infinitely many (if the two lines are exactly the same).

/the function reflect in the zero-point.
}

U4-Equations, inequalities and graphs/solving modulus inequalities

solve with linear y=R{
1.calculate it (get one intersection)
2. Get 2 intersection with refelct the graph (under the x-axis)
}slove with algebra{
|2x+3|>4

2x+3 < -4 OR 2x+3> 4
}solve with algebra with graph{
|a|>|b|
$a^2>b^2$
(factorise form)(factorize form)> 0
}solve with if x()R with graph{

|x+1|=
$x+1 \quad x \in R$ OR $-(x+1) \quad x \in R$
then calculate it (get the intersection points)
}

U4-Equations, inequalities and graphs/sketching graphs of cubic polynomials and their moduli& SCIG & SMCQE

-{ A cubic function is a function of the form

$$f(x) = ax^3 + bx^2 + cx + d(a \neq 0)$$

where a,b,c,d are constants and a is the leading coefficient.
$\text{If a} > 0\text{a} > 0, \text{as x} \to -\infty, \text{f(x)} \to -\infty\text{x} \to -\infty, \text{f(x)} \to -\infty \text{ and as x} \to +\infty, \text{f(x)} \to +\infty\text{x} \to +\infty, \text{f(x)} \to +\infty.$
$\text{f a} < 0\text{a} < 0, \text{as x} \to -\infty, \text{f(x)} \to +\infty\text{x} \to -\infty, \text{f(x)} \to +\infty \text{ and as x} \to +\infty, \text{f(x)} \to -\infty\text{x} \to +\infty, \text{f(x)} \to -\infty.$
$Roots\ \&\ factors: Can\ have\ one\ real\ root\ or\ three\ real\ roots\ (including\ repeated).$
$Use\ Factor\ Theorem\ to\ find\ a\ linear\ factor, then\ factor\ the\ quadratic\ factor.$

$Derivative: f'(x) = 3ax2 + 2bx + c\ quadratic\ used\ to\ find\ stationary\ points.$

$Sketch: Plot\ x - intercepts, y - intercept, and\ use\ sign\ of\ a\ for\ end\ behavior.$

$Modulus: y = | f(x)|\ reflects\ the\ part\ below\ xx - axis\ upwards.$
1}
In f(x)=a(x-b)(x-c)(x-d), if a is positive, then it is increases by and large. But if a is negative, then the function is decreases by and large.

{factorise:From f(x)=(simpliest factorise form) (simpliest factorise form) (simpliest factorise form)
step-by-step to calculate that.
}
About maxima and minima- global (absolute) or local(relative){
Understanding Extrema with a Cubic Example：
$\text{Consider f(x)} = x^3 - 3\text{x}, \text{domain R}.$
Local maximum: at x=−1, f(−1)=2. Nearby values are all less than 2 → local maximum.

Local minimum: at x=1, f(1)=−2. Nearby values are all greater than -2 → local minimum.

Global maximum: none, because as x→+∞, f(x)→+∞.

Global minimum: none, because as x→−∞, f(x)→−∞.

If the domain is restricted to [−2,2]:

Global maximum: at the endpoint x=2, f(2)=2 (and also at x=−1, f(−1)=2; both are tied).

Global minimum: at x=1, f(1)=−2.

Finding Local Extrema

1. Find f'(x) and solve f'(x)=0 to get stationary points.
2. Use the second derivative f''(x) to classify:
    - f''(x)<0 → local maximum
    - f''(x)>0 → local minimum
3. If f''(x)=0 use the sign change of the first derivative to determine the nature.

Finding Global Extrema on a Closed Interval [a,b]

1. Find all stationary points that lie inside the interval.
2. Evaluate ff at the endpoints a and b.
3. Compare the values at the stationary points and endpoints. The largest is the global maximum; the smallest is the global minimum.

Common Mistakes

- Confusing local with global: A cubic function on R has no global extrema, but students often mistakenly treat local extrema as global.
- Forgetting endpoints: On a closed interval, global extrema can occur at the endpoints, not just at stationary points.
- Terminology confusion: In exams, “local” or “relative” refers to local extrema; “absolute” or “global” refers to global extrema. Make sure to distinguish them.

}sketching the graph of cubic function{

$$f(x) = ax^3 + bx^2 + cx + d \; TO \; \mathrm{f(x) = a(x - p)(x - q)(x - r)}$$

1. // General direction:

$\mathrm{If\ a > 0a > 0, as\ x \to -\infty, f(x) \to -\infty x \to -\infty, f(x) \to -\infty\ and\ as\ x \to +\infty, f(x) \to +\infty x \to +\infty, f(x) \to +\infty.}$
$\mathrm{if\ a < 0a < 0, as\ x \to -\infty, f(x) \to +\infty x \to -\infty, f(x) \to +\infty\ and\ as\ x \to +\infty, f(x) \to -\infty x \to +\infty, f(x) \to -\infty.}$

2. x-intercept and y-intercept:

y-intercept： f(0)=d （simple form） OR a(−p)(−q)(−r). (factorized form）

y-intercept （root） ： Factorized form： x=p,q,r （real root） ; Simple form： Try rational roots using the factor theorem → reduce the degree by long division → solve quadratic equations

/ The image at the double root is tangent to the X-axis (does not pass through).

3. Find the first derivative and locate the stationary point (local extremum point)

$$f'(x) = 3ax^2 + 2bx + c$$

/Determine the type of extreme value using the second derivative (optional):

$$f''(x) = 6ax + 2b$$

f''(x)>0 → local minima

f''(x)<0→ local maxima

f''(x)=0→ It is necessary to use the sign change of the first derivative to determine (which might be the inflection point).

To find the y-coordinate: Substitute x_1 and x_2 into the function f(x)
4. Determine additional points (optional)

Select several x values (such as x=0, on both sides of the point), and calculate the y value to help make the connection more accurate.

5. Connect all points smoothly and pay attention to the shape of the curve

Starting from the far left, follow the direction and pass through all the intersection points and extreme points, and finally reach the far right.

The curve is smooth without sharp corners (the cubic function is differentiable everywhere).

}